



INTERNAL SECURITY

MODULE - 17

CYBER SECURITY

CYBER SECURITY POLICY, 2013

7 Build indigenous security technologies through research.

6 Create a cyber ecosystem in the country, developing effective public-private partnerships and collaborative engagements through technical and operational cooperation

5 Set up testing labs to regularly check safety of equipment being used in the country.



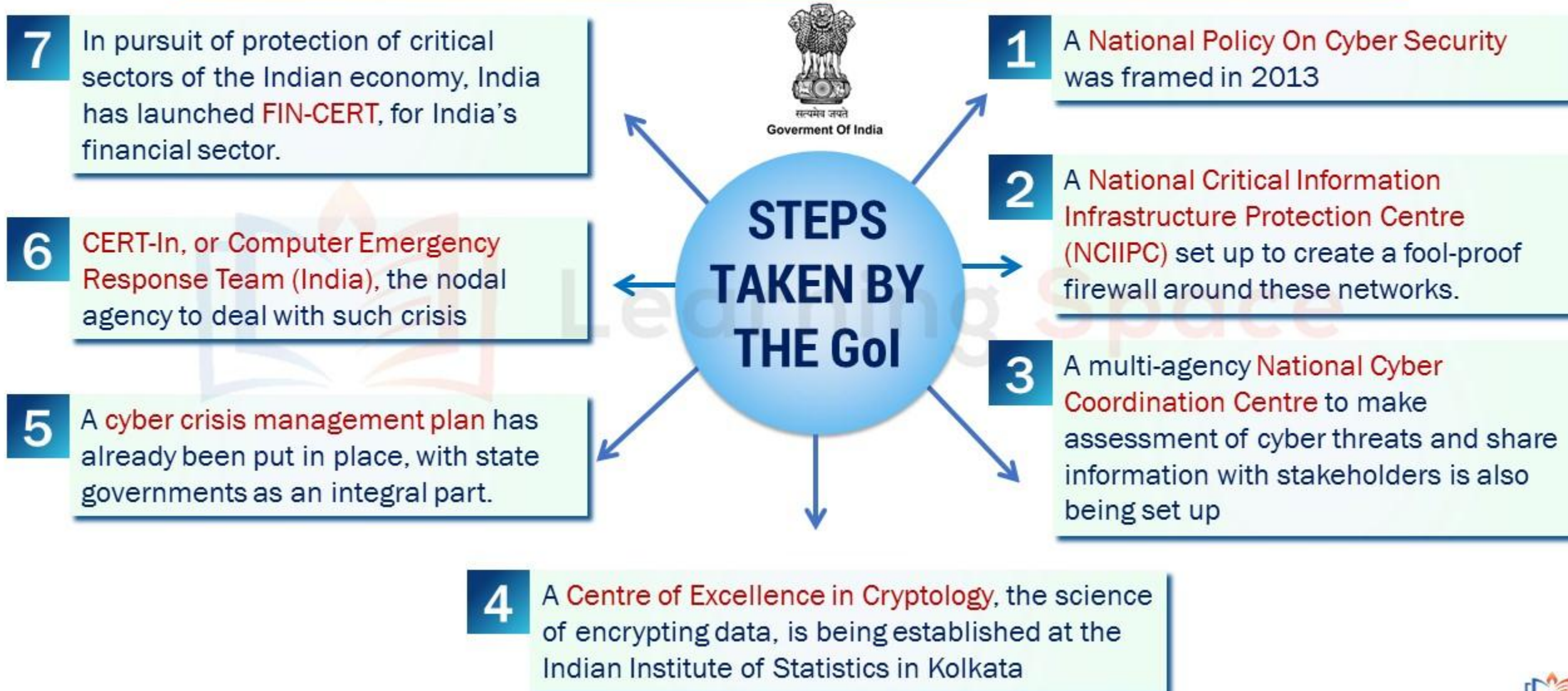
1 Set up different bodies to tackle various levels of threats, along with a national nodal agency to coordinate all cybersecurity matters.

2 Create a National Critical Information Infrastructure Protection Centre (**NCIIPC**)

3 Create a workforce of around 500,000 trained in cyber security.

4 Provide fiscal benefits to businesses to adopt best security practices.

INSTITUTIONAL MEASURES TAKEN BY GOVERNMENT



INSTITUTIONAL MEASURES TAKEN BY GOVERNMENT



STEPS TAKEN BY THE GoI

12

Under **Cyber Crime Prevention for Women and Children (CCPWC)** Scheme, Govt has released grants to States/UTs for setting up of a Cyber Forensic cum Training Laboratory and organizing capacity building programme on cyber awareness and cyber crime investigation.

8

National Cyber Security Coordinator (NCSC) under National Security Council Secretariat (NSCS) coordinates with different agencies at the national level for cyber security matters.

9

IT Act, 2000 was enacted to provide legal recognition for electronic communication, electronic commerce and cyber crimes etc. IT Act has deterrent provisions to deal with cyber threats and cyber attacks.

11

Cyber Swachhta Kendra (Botnet Cleaning and Malware Analysis Centre) has been launched for detection of malicious programs and provide free tools to remove the same.

10

Cyber security exercises are being conducted regularly to enable assessment of preparedness of organizations in Government and critical sectors.

CHALLENGES IN THE CYBER SECURITY POLICY



1

Roles and Responsibilities:

- **National Technical Research Organisation** has been entrusted with this responsibility, which doesn't come under any ministry and operates directly under Prime Minister's Office (PMO).
- The interplay between Ministry of Defence (MoD), Ministry of Home Affairs (MHA), the Armed Forces, and the intelligence agencies, both internal and external, needs to be clearly demarcated.
- **Lack of Unity of Command**- which defines who will carry out offensive cyber operations in a conflict scenario and the rules of engagement.
- **Lack of Standards and Protocols**- across the country in the cyber domain.

CHALLENGES IN THE CYBER SECURITY POLICY

2 Issues with National Cyber Security Coordinator:

- It does not have any executive power since it is not under any Ministry.
- Its faces the issue of lack of staff.



3 Coordination in Public-Private Partnership:

- **Perception of Private Agencies-** the private sector is only interested in grabbing the order, but are not serious enough in developing Indian solutions and thus do not invest much in R&D.
- **Perception of Public Agencies-** there is very little understanding of cyber security in the top echelons of the government agencies. Also, the procedures are too bureaucratic and time consuming.

CRITICAL INFRASTRUCTURE AND CRITICAL INFORMATION INFRASTRUCTURE

CI

Critical Infrastructure (CI) can be defined as those facilities, systems, or functions, whose incapacity, destruction would cause a debilitating impact on national security, governance, economy and social well-being of a nation.



CII

- Critical Information Infrastructure (CII) is broadly defined as including those information networks which are interrelated, interconnected and interdependent.
- Critical Information Infrastructure (CII) are those ICT infrastructure upon which the core functionality of critical infrastructure is dependent.

CYBER SECURITY

THREATS TO CII

INTERNAL THREAT

- It is defined as one or more individuals with the access and/or inside knowledge of a company, organisation or enterprise that would allow them to exploit the vulnerabilities of that entity's security, systems, services, products or facilities with the intent to cause harm.
- Insider betrayals causes losses due to IT sabotage, fraud and theft of confidential or proprietary information.
- This may be intentional or due to ignorance.

EXTERNAL THREAT

This threat arises from outside of the organisation, by individuals, hackers, organisations, terrorists, foreign Government agents, non-state actors, and pose risk, like crippling CII, espionage, cyber/electronic warfare, cyber terrorism, etc.

Threat may be caused by individuals, including disgruntled or former employees, rivals (industrial espionage), hackers, script kiddies, cyber criminals (organised as well as unorganised), cyber mercenaries, terrorist groups, non-state actors and hostile states.

CYBER SECURITY

FRAMEWORK TO SUPPORT CII

LEGAL FRAMEWORK

Section 70 of the IT Act 2000 takes all the measures including the research and development for the protection of CII.



INSTITUTIONAL FRAMEWORK

The government has designated the National Critical Information Infrastructure Protection Centre (NCIIPC) of National Technical Research Organisation (NTRO) as the nodal agency under Section 70A (1)



WHY WAS CII IN NEWS RECENTLY?



- 1 Recently, there were cyber-attacks on Kudankulam Nuclear Power Project (KKNPP).

- 2 National Cyber Coordination Centre received intelligence from a US-based cybersecurity firm that a 'threat actor' had breached and intercepted the "domain controllers" at the KKNPP and at ISRO.

- 3 This spyware identified as 'Dtrack' is programmed to steal data and give the hacker or the 'threat actor' complete control over all the infected devices by exposing its credentials and passwords.

- 4 This casts serious doubts on the Indian state's claims to being a legitimate power in cyberspace due to the vulnerability of its critical information infrastructure.

SIGNIFICANCE OF CRITICAL INFRASTRUCTURE PROTECTION

4 A minor disruption at one point could have a **rippling effect** across multiple infrastructures.

3 Any delays or disruptions in the functioning of these critical information systems can potentially spread across other CII, resulting in political, **economic, social or national instability.**



**CII ARE
VULNERABLE
BECAUSE...**

1 These information infrastructures are **especially vulnerable** to cyber attacks and breaches.

2 CII are **deeply interconnected** and complex by design and also **geographically dispersed.**



CYBER SECURITY

INTERNATIONAL COOPERATION IN CYBER SECURITY



1

Budapest Convention on Cybercrime, 2001:

- This convention of the Council of Europe is the only binding international instrument on this issue.
- It is the first international treaty on crimes committed via the internet and other computer networks and deals with issues such as infringements of copyright, computer-related fraud, child pornography and violations of network security.
- It provides for procedural law tools to make investigation of cybercrime and securing of e-evidence in relation to any crime more effective.
- It is supplemented by a "Protocol on Xenophobia and Racism" committed through computer systems.

CYBER SECURITY

INTERNATIONAL COOPERATION IN CYBER SECURITY



2

Global Centre for Cyber security:

- It has been launched by the World Economic Forum (WEF) with its headquarter in Geneva.
- It will function as an autonomous organisation under WEF.
- It will serve as laboratory and early-warning think tank for future cyber security scenarios and help to build a safe and secure global cyberspace.

3

“Commonwealth Cyber Declaration” at the Commonwealth Summit 2018:

- This is the world’s largest and most geographically diverse inter-governmental commitment on cyber security cooperation.



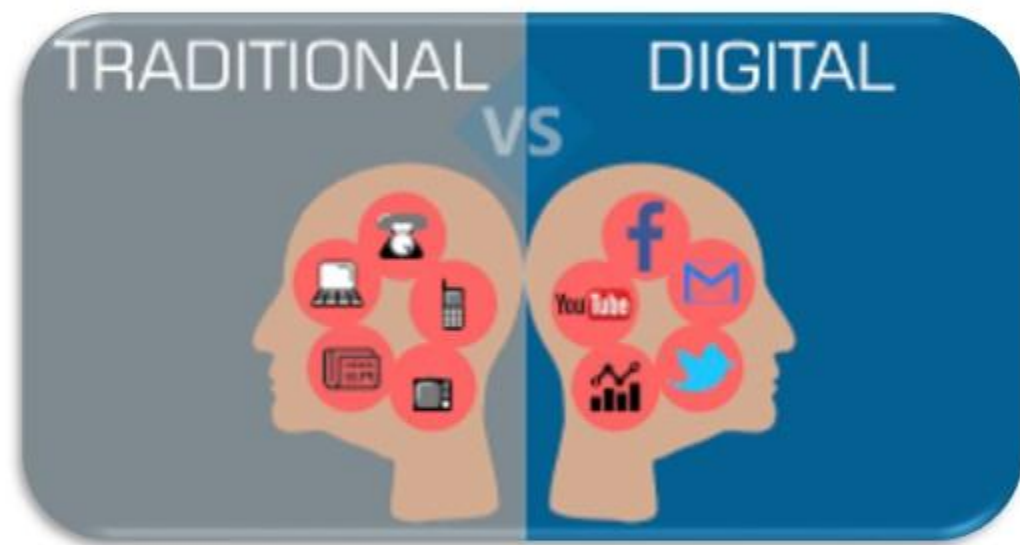
SOCIAL MEDIA



- Social media refers to internet based communication among people who create, share and exchange their ideas, photos, videos and information on virtual cyber platform.
- Its reach and popularity among people has rapidly increased over the last few years, primarily due to a sharp rise in the number of internet users and cell phone users.
- Social media includes Facebook, Twitter, You Tube, blogs, new microblogging sites etc.

SOCIAL MEDIA

HOW IS SOCIAL MEDIA DIFFERENT FROM TRADITIONAL MEDIA?



- 1 Social media rapidly passes the information to the users. It provides the real time communication of information within a very short span of time.

- 2 Traditional media and electronic media are controlled by big media houses. There is an element of monopoly in them.
- 3 But, social media is in the hands of the people. It has ensured greater transparency.
- 4 Traditional media provides one way communication. It only provides information to the users
- 5 But, social media has made our democracy more participatory. It has developed a culture of debate.

SOCIAL MEDIA

INTERNAL SECURITY THREATS DUE TO SOCIAL MEDIA



1 **Cyber terrorism:**

- Due to the convenience, affordability and broad reach of social media platforms like YouTube, Facebook and Twitter, terrorist groups have increasingly used social media to further their goals and spread their message.
- Social Media became a platform for coordination of plans of attack, communication with cells, or propaganda and information and spread of hate campaign or messages that can hurt the sentiments of people.

SOCIAL MEDIA

INTERNAL SECURITY THREATS DUE TO SOCIAL MEDIA



2 **Fraud:**

Social networking sites also invite fraudsters to take excellent opportunity to become wealthy by applying deceiver schemes.

3 **Criminal Activity and Money laundering:**

Social networking sites also pose major challenge in financial and organized crime which destabilizes the system. It creates threat to a company's security because of what employees might disclose and they are on prime target for cyber criminals.

SOCIAL MEDIA

INTERNAL SECURITY THREATS DUE TO SOCIAL MEDIA

4 To Bring Revolution:

Some countries in the world feel threatened by the fact that social media can bring the people together and thus, create a revolution. This in turn can cause political instability



5 Communal Violence and Fanning Tensions:

- Importantly, social media also seems to be playing a significant role in polarising different communities in India.
- The power of media and the process of public opinion formation in a free society had undergone radical change due to Internet and faster means of communications like SMS, Whatsapp, Viber and simplified mobile internet.

SOCIAL MEDIA

INTERNAL SECURITY THREATS DUE TO SOCIAL MEDIA

6 **Virtual Community:**

- Popular social networking websites are another means of attracting potential members and followers.
- These types of virtual communities are growing increasingly popular all over the world, especially among younger demographics.
- The chat service like Skype, which includes voice and video capabilities, has become particularly popular with terrorist cells.

- ### 7 **Hacking:**
- Hackers write or use ready-made computer programs to attack the target computer. This can make the whole country kneel without using arms and ammunition.



IS SOCIAL MEDIA POLARISING THE SOCIETY?

FAKE NEWS IN SOCIAL MEDIA



- 1 Fake news is an industry today and finds great resonance with people. Its rise corresponds with a growing distrust in the mainstream media.

- 2 Fake news has now even slipped into traditional media outlets and is often circulated by prominent individuals. This has contributed to the echo chamber phenomenon.

- 3 “Echo chamber” is a term widely used in today’s lexicon, that describes a situation where certain ideas, beliefs or data points are reinforced through repetition of a closed system that does not allow for the free movement of alternative or competing ideas or concepts.

IS SOCIAL MEDIA POLARISING THE SOCIETY?

FAKE NEWS IN SOCIAL MEDIA

- 4 Increasing polarisation on social media can be found from factors like:
- **User homophily** – users in a social system tend to bond more with ones who are similar to them than to ones who are dissimilar
 - **Algorithmic filtering** – this has created the cycle of enforcing and reinforcing belief systems and ensuring that we don't open our minds to diverse opinions.



- 5 Polarisation also happens because of:
- **Misinformation** - due to biases or human errors
 - **Disinformation** - fake news created intentionally

- 6 People seek “informed” opinions through filters only from people they trust and look for news that confirms their worldview. This results in people cultivating rigid opinions of issues.

- 7 Social media sites are more than willing to play abettors. Twitter, for example, will routinely prompt you to follow people who hold a viewpoint that is similar to yours.

SOCIAL MEDIA

EXISTING REGULATIONS AND ITS MISUSE

7 Another problem of a lot of regulatory measures is the **vagueness of language which is exploited by state agencies** to behave in a repressive way.

6 **Too stringent a policy** of policing social media could violate the individual's right to privacy.

5 These, however, have been reported to be **arbitrary**.

NEED TO REGULATE SOCIAL MEDIA



1 In India, social media platforms already come under the purview of the Information Technology (IT) Act, the '**intermediaries guidelines**' that were notified under the IT Act in 2011 and the Indian Penal Code.

2 Under existing laws, social media channels are already required to **take down content if they are directed to do so by a court or law enforcement**.

3 There are also **reporting mechanisms on these platforms**, where they exercise discretion to ascertain whether a reported post is violating community guidelines and needs to be taken down.

IS SOCIAL MEDIA POLARISING THE SOCIETY?

FINAL REMARKS



- 1 Blaming social media for every (online and offline) misconduct doesn't seem to have any logical base to stand upon
- 2 It is true that social media was used to garner support for a man who burnt a Muslim man on video in Rajasthan, but it was the same social media that was also used to raise funds for the parents of a child who was gang-raped and murdered in Kathua in Jammu and Kashmir.
- 3 This shows that social media is only just a tool, which on the hands of a user can easily do as much good as bad.

WE EXPRESS OUR
SINCERE THANKS
FOR VIEWING THIS VIDEO

Presented by



Learning Space

For Suggestions:

suggestions@learningspace.in

To Contact us:

info@learningspace.in

For Information:

info.learningspacedigital@gmail.com

Visit us at:

www.learningspacedigital.com

0866 244 44 72

98499 42299

94415 27588